

Topics: 12.4 The Cross Product; 12.5 Equations of Lines and Planes (Part I)

Please put away all electronic devices, including cell phones and calculators.

This content is protected and may not be shared, uploaded, or distributed.

1. Compute $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ for each of the following pairs of vectors:
 - (a) $\mathbf{u} = \langle -1, 1, 2 \rangle$, $\mathbf{v} = \langle 4, 5, -2 \rangle$
 - (b) $\mathbf{u} = \mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{v} = 2\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$
 - (c) $\mathbf{u} = \langle 1, 2, 3 \rangle$, $\mathbf{v} = \langle -3, 0, 1 \rangle$
 - (d) What is the significance of your answers to parts (b) and (c)?
2. Consider $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = -2\mathbf{j} + \mathbf{k}$.
 - (a) Find the cross product $\mathbf{a} \times \mathbf{b}$.
 - (b) Verify that $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{a}$.
 - (c) Find two unit vectors orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.
3. Consider the points $P(1, 2, 1)$, $Q(2, 5, 4)$, $R(6, 9, 12)$, and $S(5, 6, 9)$ in $\mathbb{R}^3$.
 - (a) Show that the vectors $\overrightarrow{PQ}$, $\overrightarrow{PR}$, and $\overrightarrow{PS}$ are coplanar.
 - (b) Find the area of the parallelogram $PQRS$.
 - (c) Find the area of triangle PQS .
4. Find the following equations of the line through the point $P(2, 2, 4)$ that is parallel to the vector $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 7\mathbf{k}$. For parts (a) and (b), use the parameter t .
 - (a) Vector equation
 - (b) Parametric equations
 - (c) Symmetric equations
5.
 - (a) Find symmetric equations of the line through the point $P(3, 4, 0)$ that is perpendicular to both vectors $2\mathbf{i} + 2\mathbf{j}$ and $\mathbf{j} + \mathbf{k}$.
 - (b) Find the point where the line from part (a) intersects the yz -plane.
6. Determine whether the lines L_1 and L_2 are parallel, intersecting, or skew.
 - (a) $L_1 : \mathbf{r}(t) = \langle -1 + 3t, 2 + 4t, 3 - 2t \rangle$, $L_2 : \frac{x-1}{2} = \frac{y}{-3} = \frac{z+1}{-3}$
 - (b) $L_1 : x = 2t, y = -3 + t, z = 5 - t$, $L_2 : x = 3 - 3s, y = 2 - \frac{3}{2}s, z = \frac{3}{2}s$
7. Find an equation of the plane through the points $A(0, 1, 2)$, $B(1, 2, 3)$, and $C(2, 3, 5)$.
8. Find an equation of the plane through the points $(0, -2, 5)$ and $(-1, 3, 1)$ that is perpendicular to the plane $2z = 5x + 4y$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Cross Product If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta.$$

Equations of a line. The line through the point (x_0, y_0, z_0) and parallel to the vector $\langle a, b, c \rangle$ has **vector equation**

$$\mathbf{r}(t) = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle.$$

The **parametric equations** for this line are

$$x = x_0 + at \quad y = y_0 + bt \quad z = z_0 + ct.$$

We can eliminate t to obtain the **symmetric equations**

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c},$$

provided a , b , and c are all nonzero. If a component of the direction vector is zero, the corresponding coordinate is constant.

Equations of a plane. The plane through (x_0, y_0, z_0) with normal vector $\langle a, b, c \rangle$ is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

By collecting terms, we can rewrite this equation in the form

$$ax + by + cz = d$$

where $d = ax_0 + by_0 + cz_0$ is a constant. Two planes are parallel if their normal vectors are parallel, and they are perpendicular if their normal vectors are orthogonal.

Suggested Textbook Problems

Section 12.4: 1-22, 27-38, 43-46

Section 12.5: 1-72